

User Interface Design of a Functional CQI Module in OBE Nexus: An Iterative Design Approach

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Abstract

Bangladesh adopted Outcome-Based Education a decade ago, receiving Washington Accord accreditation in 2023. The accreditation requirements from the Bangladesh Accreditation Council (BAC) and the University Grants Commission (UGC) forced almost all the universities to practice OBE. Continuous Quality Improvement (CQI) in OBE requires tracking and assessing an overwhelming amount of information and activities. This paper presents a user-centered design approach towards a process-oriented user interface. Through three rounds of iterative feedback and redesign, the design requirements to improve the accessibility and user experience of the system were identified and discussed. Satisfactory results in the user evaluation affirmed the design philosophy.

Keywords

Outcome-Based Education, Continuous Quality Improvement, User-Centered Design, Iterative Design, Process oriented workflow

ACM Reference Format:

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1 Introduction

The tertiary level education system in Bangladesh is transitioning from the traditional education system to the Outcome-Based Education (OBE) system, since almost a decade. This transition ensures that the graduates possess the practical and theoretical knowledge necessary for the workforce [37]. Unlike the traditional education system, OBE focuses on "Outcomes" as a Key Performance Indicator (KPI) in the education system. Outcomes measured at different levels are targeted for improvement; this process is called the Continuous Quality Improvement (CQI). The CQI processes involve feedback, tracking and monitoring of OBE activities, such as, calculation of attainment of Course Outcomes (CO Attainment). From a process engineering perspective, OBE introduces new processes

(e.g., CQI) to the organizational operational architecture. In the context of CQI, institutions face an ever-growing challenge: how to efficiently track, assess, and improve the outcomes in alignment with the accreditation standards.

The Board of Accreditation for Engineering and Technical Education (BAETE) provides accreditation for engineering programs in Bangladesh, aligning with Washington Accord standards [29]. Striving to enhancing the global mobility and quality of engineering professionals, BAETE's accreditation requirements necessitates the standards of certain OBE practices and processes. The processes are described in comprehensive detail in the website of BAETE [4] and in their other publications, such as the BAETE Manual [5].

It is observed that the CQI process is typically led by a senior faculty member. Most of the CQI tasks are performed manually relying on human memory and organizational culture of documentation. The course-level CQI activities (e.g., instructor feedback, student evaluation) are performed routinely at course completion, while at Program and Department Levels CQI takes place every 3-5 years. CQI necessitate analysis of an overwhelming amount of data, even for smaller organizations, proving the insufficiency of excel-based approaches. On the other hand, access to historical information is heavily dependent on organizational memory based on human experience, as well as on the documentation, filing and labeling culture within the organization, often causing distractions and increased cognitive burden. Even in a software environment, presentation of a large amount of qualitative and quantitative data meaningful to the user role remains as a design challenge.

In the spirit of facilitating the complex CQI processes for different stakeholders, we address the following research questions:

RQ1. *How can the CQI process be facilitated with technology in the context of the undergraduate engineering education in Bangladesh?*

From a process engineering perspective, we identified the data object references involved in different levels of the CQI process from semi-structured interviews with several participants, who are experienced in the CQI process. Modeling the CQI data flow to the processes identified the data objects relevant to the process and stakeholder. We argue that technological means can be utilized to track and assess the KPIs relevant to CQI process, within the OBE system.

RQ2. *How the relevant CQI information could be made available to the stakeholders in a meaningful way?*

CQI/SAC members are required to process a large amount of information of different types. Considering volume and variety, filtering out the data objects irrelevant to the role or to the process is cumbersome without any technological assistance. Presence of



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both qualitative and quantitative data objects increase the cognitive load to the user. Higher level CQI processes require aggregation of information meaningful to the role.

Hence the contributions of this article are twofold. *First*, we identified the data objects relevant to the CQI process, from a process engineering perspective. We modeled the flow of information from the data objects to the CQI processes for the Course Level, Program level and Department level, and argued that such systems could act as a data driven decision support system for OBE. *Second*, we described the iterative process of design and development of the CQI interface, and discussed the key design objectives for a functional UI. Rounds of iterative refinement revealed the utility of simple design choices for meaningful representation of relevant information for the CQI process. Integrating the user feedback, we aimed to create a user-centered solution for practical CQI needs. Our research seeks to address the gap in current literature by presenting a CQI module that guides CQI process presenting certain information relevant and meaningful to the user role.

2 CQI Process Overview

Continuous Quality Improvement (CQI) is one of the fundamental cornerstones [1] of the Outcome Based Education (OBE) where the goal is to improve the quality of education. A functional CQI process involves several subprocesses at different levels of the education system. According to the BAETE guidelines, CQI process can be conducted on Course level, Program level and Department level to track the attainments of Course Outcomes (CO), Program Outcomes (PO) and Program Educational Objectives (PEO). Each of the processes require careful evaluation of relevant historical data to write an action plan for improvement, which needs to be propagated to the concerned stakeholder for implementation. After the implementation of the action plans, in the next CQI cycle, the impacts of the previous actions (according to the plan) are analyzed in the form of historical data and the CQI process is carried out again, closing the loop. For example, at the course level, the focus is on improving the attainment of the Course Outcomes (CO Attainment). After the completion of the each course, the course instructor writes an improvement plan considering several data objects, e.g., CO Attainment, Student Feedback, Course Level recommendations from previous CQI cycle and many more.

The quantitative data objects are typically aggregated using spreadsheet-based solutions, and based on the organizational internal business logic (e.g., CO to PO and PO to PEO mappings) higher level data objects are created. The aggregation of the outcomes at different levels are shown in figure1 depicting the creation higher level data objects.

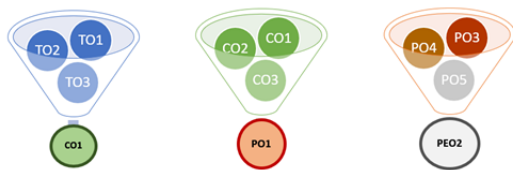


Figure 1: Aggregation of Outcomes in OBE.

3 Challenges in Implementing CQI

While quantitative data can be computationally aggregated, qualitative data requires human intelligence [42]. The lack of a software tool for CQI to track and analyze outcomes in a scalable and efficient manner weakens the OBE process [20]. The qualitative data are often difficult to track manually without proper filing and documentation culture. BAETE's rigorous guidelines and standards for CQI process could be facilitated with modern approaches combining Natural Language Processing (NLP) [12] and Machine Learning (ML) to assist in qualitative data analysis [31].

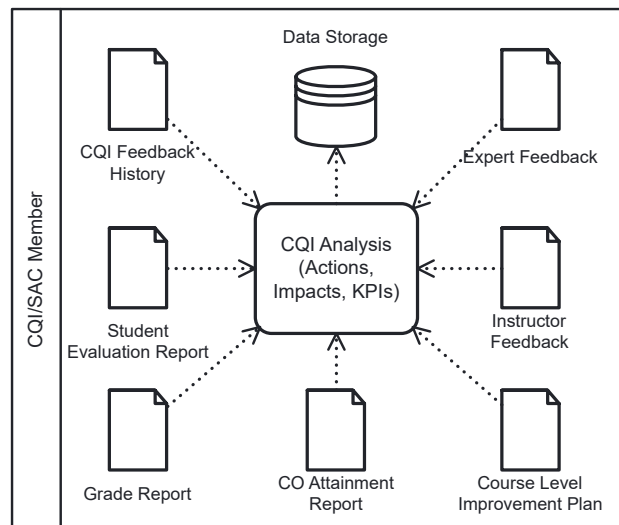


Figure 2: Data Flow in Course Level CQI

Figure 2 describes the information flow for the course-level CQI, where the course instructor works with several data objects. Quantitative data objects like CO Attainment Report and Grade Report could be generated automatically at course completion. Course-level CQI requires the teacher to provide qualitative feedback and write an improvement plan considering the data objects accessible to him. The efficacy of the improvement plan necessitates comparing the CO Attainment levels with historical data, which are not readily accessible in spreadsheet-based approaches.

In the role of CQI member, the course-level CQI process also takes input from the "Student Evaluation" and "Expert Feedback" processes. Recommendations and historical information from the previous CQI cycles are also analyzed to benchmark the performance of the CQI process. CQI members often need to remember certain key information while assessing the KPIs for the CQI process. Figure 2 shows that handling this overwhelming amount of data even for smaller institutions is quite challenging without any help of technological means. Manual data tracking is prone to errors and inefficiencies. The growing volume of data related to student outcomes, faculty performance, and course evaluations makes it increasingly difficult to manage without automated systems [15]. Commercial solutions for OBE and CQI management do exist, but they are often expensive and not well-suited to the specific needs of Bangladeshi universities [17].

In consultation with the BAETE Accreditation Manual [5] and experiences from institutional OBE practices, we modeled the CQI data flow at different levels of the OBE framework. Representing the data objects flowing towards a process, helped us visualize the relevant information, as needed by the CQI members. The PO level CQI process is depicted in Figure 3.

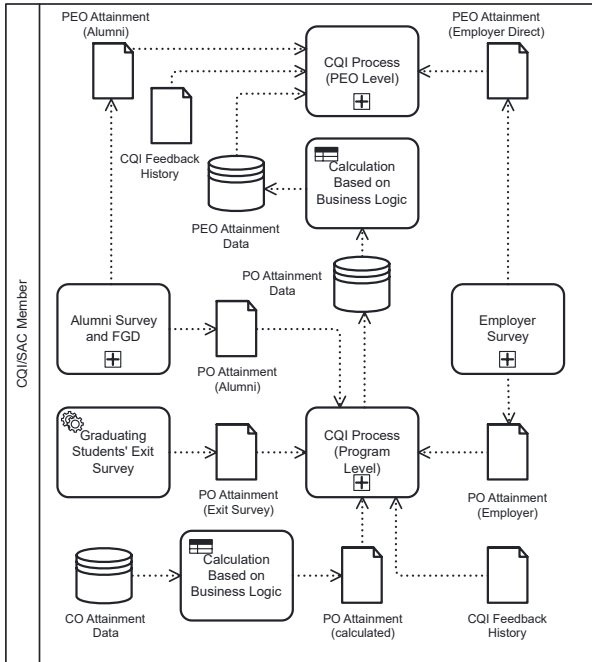


Figure 3: Data flow in Program and Department Level CQI

Higher level CQI processes require interaction among the stakeholders, and proper documentation. This additional administrative burden associated with the CQI process is cumbersome and often unpaid. Levels of OBE adoption and styles of practice vary among institutions, and executed differently. Smaller-sized institutions are being deprived of the full benefits of OBE due to the high subscription fees of the commercial solutions.

4 Related Work

The Continuous Quality Improvement (CQI) process varies across institutions and has been widely discussed in the academic literature. The role of technology in improving CQI processes was explored by Fernandes in [16], emphasizing the importance of data management systems in academic environments. The authors argued that manual or spreadsheet-based tracking is inadequate for the complexity and volume of data required by Outcome-Based Education (OBE). Similarly, Boskovic et al. [7] highlighted the role of technology in improving the learning outcomes in higher education, focusing on OBE's impact on institutional quality assurance. Institutions with budget constraints are often deprived of these technological solutions due to high development or subscription costs. Commercial solutions often require extensive training, thereby widening the digital divide [17]. Our paper builds on these works

by offering a cost-effective, user-centered alternative specifically tailored to the Bangladeshi context.

In-house development of CQI systems requires specific institutional commitments. Hasan [20] advocated for in-house development due to the flexibility in aligning with institutional activities. This aligns with the findings of Agir et al. [2], who emphasized the importance of developing Outcome-Based Assessment (OBA) systems to continuously improve educational outcomes through systematic data collection and analysis. Process-oriented workflows in software systems can streamline CQI activities in academic environments, as institutions require tailored solutions for specific workflows and data management [28]. This was echoed by Piyasena et al. [32], who discussed the challenges of OBE implementation, particularly in developing robust assessment tools and processes for continuous improvement.

Moreover, the absence of effective data visualization has been highlighted as a significant barrier to informed decision-making by educators. Mustafa et al. [28] recommended advanced visualizations that integrate qualitative and quantitative data to present information in a way that is meaningful for decision-makers. Tanabe [38] further explored how technology could assist in automating OBE processes, particularly in specialized fields such as medical education, by tracking performance against learning outcomes using tailored visualization tools.

Despite the availability of commercial OBE solutions, these systems often do not fully address the specific needs of the Bangladeshi context. Institutions in developing countries frequently struggle with the high costs and complexities of these systems. For instance, Chowdhury et al. [11] discussed the limitations of commercial solutions in automating assessment and evaluation in OBE frameworks, proposing in-house systems as more adaptable and cost-effective alternatives. Similarly, Husna et al. [21] focused on improving student competence through OBE-based learning management, highlighting the need for contextually relevant technological solutions to continuously refine the processes and adapt to the evolving requirements of accreditation bodies in the modern educational frameworks of OBE.

No existing commercial OBE solutions fully address the specific needs for the Bangladeshi context. Commercial solutions are often criticized in terms of costs and accessibility, especially in developing countries like Bangladesh, where budget constraints limit the adoption of sophisticated software systems [24]. Our work contributes to the literature by offering a user-centered design and a process-oriented system that is both cost-effective and scalable, addressing the specific challenges identified in these recent studies.

5 Methodology

In this study, we employed a user-centered design (UCD) methodology [9] to develop a CQI interface to facilitate the CQI process in Bangladeshi universities practicing OBE. The iterative design approach ensured actual user feedback and addressed practical CQI requirements. This approach aimed to address the gaps identified in the existing solutions, particularly in terms of usability and adaptability to local institutional needs [17].

Purposive sampling allowed us to focus on a particular subset of the population that are most relevant to the research question [30].

Faculty members from the CSE and EEE departments of undergraduate level with varying levels of exposure to OBE were chosen to provide diverse perspectives. This method deemed necessary to gather nuanced feedback on the usability and workflows [41]. Convenience sampling could have been easier to conduct logistically [14] because of the participants' ease of access and availability, but it would not have guaranteed that the participants would possess the necessary background to provide meaningful feedback. Snowball sampling would could have limited the diversity of perspectives.

Several potential participants were selected based on their level of involvements with the OBE practice. Participants were reached out to over phone and email, and ten faculty members participated in the semi-structured interviews and focus group discussions. Demographics of the participants mentioned in Table 1 demonstrates the participants' varying levels of experience (mean: 5.6 years, SD: 2.25 years) and roles.

5.1 Data Collection Methods

The user evaluation sessions were conducted in the form of semi-structured interviews and focus group discussions after prototype testing. We provided interview guides and informed consent forms prior to the interview sessions. Semi-structured interviews allowed more freedom to discuss the challenges and institutional perspective besides providing insights into the potential design solutions. Focus group discussion allowed collaborative evaluation of the UI, covering diverse viewpoints. The participants interaction with the prototype of the CQI interface enabled them to provide real feedback on usability of the system. All sessions took place on Google Meet for convenience and efficient documentation.

5.2 Iterative Design and Refinement Process

The design process was iterative, consisting of multiple cycles of feedback and refinement. Our workflow approach is depicted in Figure 4. The iterative process of user evaluation, note-taking, and thematic categorization was repeated, allowing us to continually adjust the design choices to align with user needs. This approach aligns with the principles of User-Centered Design (UCD), emphasizing continuous user involvement and iterative refinement [18].

our first design was on paper with hand-drawn sketches, that served as a foundation for our initial discussions with the participants. As a part of our professional engagements, we analyzed several commercial OBE solutions, that gave us a head-start in preparing the sketches depicted in figure 5.

Feedback from P1 highlighted the institutional challenges and expectations from the system, guiding initial design requirements. Identifying the design requirements involved thematic analysis of the interview data, which included note taking, repeated playbacks of the recording, theme extraction and categorization. The identified keywords were translated into design requirements and refinements. Interviews with P2 and P3 led to the second iteration of the design with the *Bootstrap* CSS framework, as depicted in figure 6, where key design objectives such as a process-oriented navigation and cognitive load reduction were identified. As a design response, we rearranged the menu items in the left sidebar, added

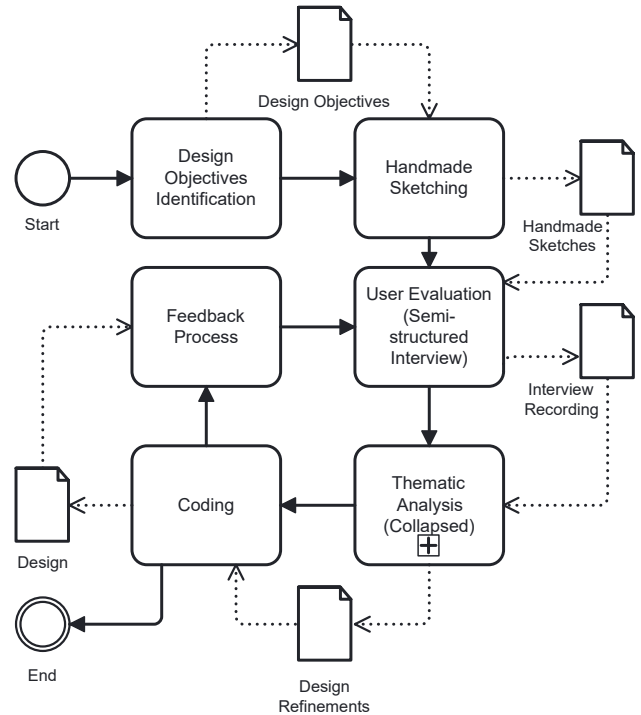


Figure 4: Iterative Feedback and Design Process

contextual icons, introduced collapsible sections, and included key-based searching. Role-based interfaces were introduced to reduce cognitive load.

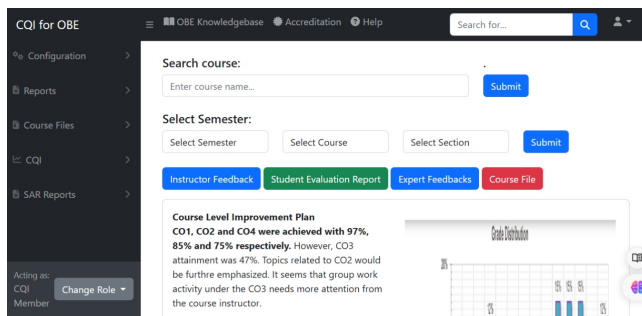
Focus group discussion with P3–P8 was preceded by prototype testing session, where the participants were given access to use the system before the discussion. Insights from the FGD sessions were categorized as accessibility enhancements, contextual help materials, open access to OBE Knowledgebase. As a design response, we resorted to *Javascript* frameworks to incorporate the changes



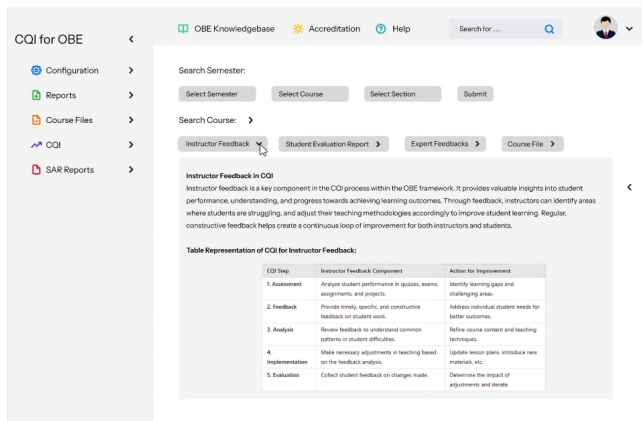
Figure 5: Initial Conceptual Sketches of the UI

Table 1: Participants' Profile

Participant	Qualification	Designation, Department	Institution	OBE Exposure since (year)
P1	Ph.D.	Associate Professor, CSE	O1	2016 (BAETE Evaluator, CQI Member)
P2	Ph.D.	Associate Professor, CSE	O1	2019
P3	Ph.D.	Associate Professor, CSE	O1	2020
P4	Ph.D.	Assistant Professor, EEE	O2	2016
P5	Ph.D.	Assistant Professor, EEE	O3	2019
P6	M.Sc.	Senior Lecturer, CSE	O1	2017
P7	M.Sc.	Senior Lecturer, CSE	O1	2017
P8	M.Sc.	Lecturer, CSE	O1	2022
P9	Ph.D.	Associate Professor, EEE	O4	2016 (BAETE Evaluator, SAC Member)
P10	Ph.D.	Assistant Professor, EEE	O5	2022

**Figure 6: CQI interface (second round of evaluation)**

producing the third iteration of the design, as depicted in figure 7. In the final round of user evaluation, P9 and P10 affirmed the design choices and no critical design issues were identified. We therefore finalized the design of the CQI interface.

**Figure 7: CQI interface (third round of evaluation)**

5.3 Data Analysis Methods

For the analysis of the interview data, we resorted to thematic analysis [8] to interpret the qualitative data in terms of design objectives.

Note-Taking and Recording: The interview and FGD sessions were recorded with informed consent, and notes were taken to capture key points without full transcription. Further refinements of the notes involved repeated playbacks of the recordings without the resource-intensive process of transcription, as argued in [19]. The decision not to fully transcribe the interviews is supported by methodological considerations in qualitative research. Halcomb and Davidson argued that detailed note-taking combined with iterative listening can be sufficient for thematic analysis. This approach is practical and maintains the integrity of the data collection and analysis process.

Thematic Coding: We categorized the suggestions and feedback into themes such as *Accessibility Challenges*, *Information Overload*, *Generalized Interface*, *Process Overview*, and *Help Materials*. This approach allowed us to filter-out the repeated suggestions and identify design refinements.

5.4 Ethical Considerations

Ethical guidelines for human subjects research were strictly followed. We adhered to the ethical guidelines for research involving human subjects. We established rapport with the potential participants over phone or email prior to the feedback sessions.

- **Consent:** Informed consent was obtained from all participants, who were briefed on the study's purpose and their rights.
- **Confidentiality:** Participant anonymity was maintained by using identifiers (e.g., P1, P2) instead of names.
- **Data Handling:** Recorded sessions and notes were securely stored and used solely for research purposes.

6 Findings and Lessons Learned

In this section we describe the key findings and present arguments on the validity of the design choices through repeated cycles of user feedback, as depicted in Figure 4. Several authors of this paper are engineering academicians giving us a head-start in research problem formulation and plan the workflow. Our initial designs were handmade sketches with pen and paper depicted in Figure 5.

In our primary interview sessions, P1 highlighted the institutional drivers and barriers of technology adoption for OBE. P1 identified process automation, efficient documentation and tracking the outcomes as drivers, and high subscription costs and institutional

culture, data variety as barriers. We also asked P1 about his expectations and design recommendations to facilitate the workflow within the system. P1 emphasized on a simplistic process-oriented design, highlighting the variations in OBE practices in different organizations. We identified key design requirements from the thematic analysis of P1's responses.

P1's insights were combined with the feedback from P2 and P3, which we collected in a similar fashion. P2 advocated for the CQI system to be a *data-driven decision support system*, providing access to historical information for the users of the system. P2 also suggested strategies for in-house development. On the other hand, P3 emphasized on increased tracking of the KPIs (e.g., attainments of COs and POs) due to difficulties manual tracking that relies on human memory. In most of the cases, P3's recommendations were identical with the previous participants, except P3 suggested the inclusion of collapsible sections, and key-based searching advanced searching, adding informative resources in the top navbar.

We performed thematic analysis on the raw data and translated them into an initial design depicted in Figure 6. In this round, we emphasized on the CQI process oriented navigation within the system. For example, program level data objects are not considered for the course-level CQI process. Furthermore, a course instructor concerned with course-level data objects, may find program level data objects irrelevant. As P2 said, *"Course instructors are primarily concerned with tracking course-level outcomes, while the CQI/SAC committee members would require access to broader program level data."*

As design response, we rearranged the left sidebar according to the CQI processes, and introduced a role-based access for the CQI processes. With these design choices, the interface remains simple for a course teacher performing course-level CQI activities, or for a CQI/SAC member performing CQI activities at the Program-level or higher.

We presented the newly designed interface to a set of participants from different institutions (P3, P4, P5, P6, P7, P8) in a focus group discussion session, demonstrating the user interface and the workflows. Based on their evaluations, a number of crucial design improvements were identified and implemented in the UI afterwards. Design refinements were performed once again.

At the final stage, we presented the interface to P9 and P10. P9 having experience with the BAETE accreditation and the CQI process, appreciated the attempt and affirmed the design choices. P9 also expressed interest to provide further feedback on the future designs and user experience. P9 suggested some accessibility improvements like adjustable font size, meaningful aggregation and visualization of data. Our final participant P10 is the least experienced in OBE practices. P10's institution is yet to start the practice of OBE, and currently in the planning and training stage. P10 having attended a number of workshops and OBE training programs, was well-aware of the CQI processes, commented on several features that require integration with the institutional automation system. P10 argued about the standardization of the educational data exchange format, for future interoperability and scalability. We limit the discussion in this paper up to the design choices part and keep the several insights from the interview data for future research.

From the user feedback, we identified the critical design objectives and refinements related to the research questions, and implemented the following design choices to present the information to the potential users in a meaningful way. We summarize and justify them as follows:

Responsive design: A modern interface responsive to various screen sizes. We used CSS and Javascript frameworks to ensure the responsiveness of the UI.

Process oriented navigation: Based on the CQI workflows and the data flows, we organized the menu items related to the CQI process. Users also appreciated the inclusion of a process overview page at the beginning of the workflow, since higher level CQI processes run after 3 to 5 years in Bangladesh.

Reduction of cognitive load: The interface should not clutter the view, collapsible accordions could be utilized to hide/show the desired information on the screen, keeping the information readily available just one click away, preventing long scrolling. We placed the contextual help menu hidden by default for the experienced user to reduce cognitive load. This part deserves further research.

Role-based Access: To ensure data privacy, access to data objects and workflows would be restricted among the designated users only, making the user interface intuitive. Each user will only see the data objects relevant to his role and the process.

Key based searching: Experienced users of the system prefers a key-based searching (e.g., 2024.1.EEE101.3 is a key that corresponds to the section 3 of EEE101 course, offered in the 1st semester of the year 2024). This also makes the searching easier while with mouse it would require at least four clicks.

Access to Knowledgebase: Access to the knowledgebase of OBE, Accreditation and relevant items are offered to all visitors by placing them at the top navbar. All the participants agreed that the knowledgebase could be made freely available for anyone interested.

Accessibility Features: Considering the potential users of the CQI system would be senior faculty members, certain accessibility features like adjustable font size, high contrast, contextual icons and colors need to be used.

We conclude the reporting of our findings with a positive response from P1 about the potential of the system. P1 mentioned that, *OBE was perceived to be very tough at the beginning. But the truth is, OBE can be customized within the guidelines of BAETE accreditation requirements. Different types of assessments could be implemented. Let's try to be adaptive. Incremental development with an Agile Approach could be adopted. Starting with the simplest module. The requirements could be changed dramatically, therefore, the development needs to be adaptive.*

7 Discussion

We demonstrated that technological assistance could be utilized to facilitate the CQI process. We took a top-down approach to address the issue of variations in institutional practices of OBE. From an organizational perspective, CQI is another aggregated business process that depends on several other processes. Due to the variety and volume of data, the CQI process becomes quite challenging without technological means. To answer the RQ1 from a process engineering perspective, we first modeled the data flows associated

to the CQI processes at various levels, which clarified what to show on the screen, and to whom. Adopting a process oriented interface can greatly mobilize the CQI process at different levels.

The main challenge to answer the RQ2 is the amount of qualitative information to be presented, in a meaningful way. Quantitative data can be easily aggregated with mathematical operations, however, qualitative data aggregation require advanced level features, that may not be contextually correct. Unless NLP and ML techniques are employed, the CQI member must rely on his human memory. Paper-based documentation (e.g., course files, meeting minutes, improvement plans) puts additional cognitive burden, while simple design techniques could be utilized to reduce the cognitive load on human. Furthermore, the overwhelming amount of information could often make the user lose focus to another data object irrelevant to the CQI process. With simple collapsible accordion sections, we argue that presenting a large amount of relevant qualitative information could be placed only one click away, offering a variety of information meaningful to his role.

This article focuses on a process-oriented user interface (UI) design, while previous studies emphasize automation of outcomes assessment and integration of analytical tools. Liew and Saad [27, 33] discussed automation tools to support the CQI process.

Another study engaged administrators and advisory boards to ensure the effective CQI implementation [25, 43] while ours involved only faculty members as the potential users. Earlier studies also present quantitative techniques such as statistical methods to measure program outcomes, as demonstrated by Tanjong in [39], while in this paper we present them to the user to qualitatively assess the outcomes. Finally, we advocate for flexibility through role-based access to reduce cognitive load, while another study in [43] relies a holistic approach like Total Quality Management (TQM) for process efficiency and structure.

UCD and Iterative Design has been widely adopted in Healthcare and Education. A telerehabilitation platform for post-COVID-19 recovery requiring accessibility for different age groups [34], a diabetes management app requiring continuous feedback emphasizing the needs of the underserved [6], a telehealth-enhanced cardiac rehabilitation program through multiple rounds of testing and refining the prototype [13], a serious educational game on phonological awareness in children [40], a tangible interface for dyslexic children [22], all of these cases utilized UCD and iterative cycles to improve usability.

Authors of [10] explored spreadsheet-based approaches to track COs and POs, that are quantitative. Shin investigated the perceptions of the professors on CQI at a university [35]. Data visualization tools were proposed to enhance the educational information systems in [36]. In [3], the authors evaluated UCD in software development, and Le [26] extended UCD practices by incorporating participatory design in automotive applications in the context of Human-Machine Interaction (HMI).

In comparison, our work integrates UCD and iterative UI design for the CQI processes in OBE. We utilized simple design choices to create a process-oriented UI to streamline CQI workflows based on our mapped the data flows and stakeholder roles to supporting continuous improvement in educational outcomes. Our approach addresses a gap in the literature by applying UCD and iterative

Table 2: Comparison with Existing Literature

Aspect	Our Contribution vs. Existing Literature
UCD Methodology & Iterative Design	UCD and iterative design are widely applied in [6, 13, 34], but our work integrates these specifically for CQI in OBE.
Focus on OBE/CQI	Studies address quality improvement in education [10, 35], but none directly focus on CQI in OBE.
Role-Based Access & Process-Oriented Interface	Existing works [3, 23] do not address role-based access or process-oriented interfaces for CQI tasks like ours.
Educational Process Emphasis & Tailored for OBE Context	Our work specifically addresses educational CQI processes in OBE, an area not covered by other general UCD approaches [22, 36].
Comprehensive Data Flow & Stakeholder Mapping	Our work includes detailed process flows and role-based mapping, enhancing user experience in educational CQI, unlike others that focus on narrower UCD applications [26].
Unique Contribution	We uniquely combine UCD, iterative design, role-based access, and process-oriented CQI tasks specifically tailored to OBE.

design to assist complex educational processes, with enhanced usability.

8 Limitations and Future Work

In this paper, we limited ourselves to the UI design part only. However, several future works remain in this context. Aggregation of several data objects into a single view remains a design challenge based on the specific context. Apart from the CQI processes, similar studies could be conducted on other OBE processes like Course File archiving, Documentation, Constructive alignments tracking. Modeling complex processes like data aggregation and visualization could further aid the interface design. After the full implementation, heuristic evaluations and user acceptance tests could be performed and analyzed based on the theoretical HCI frameworks. Intelligent recommendations could be introduced utilizing NLP techniques.

Identifying the patterns in the workflow could facilitate the design process even further, enabling the developer to reuse a module. From the interview data, it was evident there is a need of exploratory studies in the context of engineering education in Bangladesh investigating the institutional perspective on addressing the challenges of effective OBE implementation.

This study was highly focused on the undergraduate engineering education only. However, similar approach could be taken in the other domains too. Existing literature review indicates the health-care sector could be a potential use-case of the CQI system. Testing the system with broader user groups under a different context could reveal interesting insights into the design choices.

9 Conclusion

We discussed the challenges of implementing Continuous Quality Improvement (CQI) processes within Outcome-Based Education

(OBE) systems in the context of undergraduate engineering education in Bangladesh. Understanding the overwhelming complexities of the CQI process and data management requirements, we designed a functional user interface to facilitate the CQI process effectively. Faculty members who are involved in the OBE practices were engaged in the iterative design and feedback process with a user-centered design (UCD) methodology. Through multiple rounds of user evaluations, we identified critical design requirements and refined our interface to meet the practical needs of the stakeholders.

Our findings highlight that a process-oriented interface, coupled with role-based access control, significantly enhances the usability and efficiency of the CQI module. We reduced the cognitive load to the user and streamlined the user experience by making relevant information available within a single click, organizing navigation according to CQI workflows and restricting access to relevant data based on user roles. Accessibility features such as adjustable font sizes and high-contrast themes made the interface more user-friendly, particularly for senior faculty members. Aggregating qualitative data remains a challenge for future enhancements to facilitate the analysis of qualitative inputs such as feedback and improvement plans.

This study contributes to the field by demonstrating how technological solutions can be tailored to support complex educational processes like CQI in OBE frameworks. Our process-oriented approach and emphasis on user feedback ensured that the interface aligns with actual workflows and meets the specific needs of educators in the Bangladeshi context. In conclusion, the development of a functional CQI module with a user-centered, process-oriented design holds significant potential for enhancing the implementation of OBE in educational institutions. By facilitating efficient data management and reducing administrative burdens, such tools can contribute to continuous improvement in educational quality and alignment with international accreditation standards.

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